

Hazrat Usman

Ichra, Lahore

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Professional Summary — As an AI Engineer, I specialize in designing, developing, and deploying intelligent systems. My expertise spans the complete AI lifecycle, from foundational supervised learning projects to complex AI automation and the development of advanced models in cutting-edge domains. My practical experience includes impactful roles such as an AI Computer Vision Intern at NCAI (UET Peshawar) and KPITB, where I leveraged my robust software engineering and AI background to build and deploy real-world, production-ready AI solutions.

Education

Cecos University Peshawar 2024 – 2026
MS Software Engineering (Research: Flood Detection via SAR/Multispectral Data)

Cecos University Peshawar 2020 – 2024
BS Software Engineering (FYP: Honey Adulteration Detection Through AI)

Technical Skills

- **Programming Languages:** Python, SQL
- **AI & Machine Learning:** Pandas, Matplotlib, Scikit-learn, TensorFlow, NumPy
- **Computer Vision:** YOLO, Vision Transformers (ViT), Swin Transformer, OpenCV, CUDA
- **Generative AI & NLP:** LangChain, n8n, RAG, Groq, Pinecone, Gradio
- **MLOps & Deployment:** Azure ML, AWS, Flask, Streamlit, Git

Work Experience

National Center of Artificial Intelligence (NCAI) June 2025 – Nov 2025
AI Computer Vision Intern Onsite, Peshawar

- Developed and deployed a hyperspectral **Honey Adulteration Detection** system using classical ML (SVM, XGBoost) and a Flask API.
- Engineered a **Traffic Management System** by applying **transfer learning** and **model fine-tuning** to train three distinct YOLOv8 models (Vehicle Detection, Helmet Detection, License Plate/OCR).
- **Transformer-based Object Detection** – Applied **Swin Transformer** with YOLOv8 backbone, achieving improved detection accuracy compared to baseline CNN-based models.
- Built an end-to-end **RAG-based chatbot** using Pinecone vector search and the Groq LLM, enabling intelligent, document-aware Q&A via a Gradio interface.
- Automated and orchestrated **ML pipelines** and model deployments using AWS SageMaker.

Khyber Pakhtunkhwa Information Technology Board (KPITB) May 2025 – Nov 2025
Artificial Intelligence Intern Hybrid, Peshawar

- Built, trained, and optimized supervised ML models (e.g., Random Forest, XGBoost, SVM) on tabular (CSV) datasets, applying comprehensive cross-validation and hyperparameter tuning.
- Performed comprehensive data preprocessing, feature engineering, and hyperparameter tuning to optimize model accuracy.
- Developed and deployed interactive web applications and dashboards using Streamlit, allowing non-technical stakeholders to interact with and consume model results.
- Built and deployed lightweight REST APIs using Flask to serve model predictions.

Publications and Project Presentations

ICTIS UET Peshawar April 2024
ICT Pasha Awards August 2024
DATAFEST 2024 Pakistan China Friendship Centre Islamabad September 2024

Certifications

Introduction to AI and Machine Learning on Google Cloud – Coursera
Programming for Everybody (Getting Started with Python) – Coursera
AWS Cloud Technical Essentials – Coursera
Introduction to Devops - Coursera
Deep Learning Specialization – Coursera